

Report: Optimising NYC Taxi Operations

Include your visualisations, analysis, results, insights, and outcomes. Explain your methodology and approach to the tasks. Add your conclusions to the sections.

1. Data Preparation

1.1. Loading the dataset

New York City is a living organism, where the iconic yellow taxi serves as the lifeblood of its transit arteries. We begin to prepare for the analysis by importing certain Python libraries such as – pandas, numpy, matplotlib, and seaborn. These libraries will help us with manipulating and visualizing data. Then, we load the data from the trip_records folder using the multiple Parquet files. In my case, I have saved all my files on my Github account.

URL: <https://github.com/mayuresh-tungare/eda-assignment/tree/main>

1.1.1. Sample the data and combine the files

When dealing with millions of records, "more data" is not always "better data" if it impedes efficiency of processing. We cannot process the entirety of NYC's 2023 taxi records simultaneously as that is computationally prohibitive, especially given the computers that I have. Therefore, I had to use a strategic sampling method to make the dataset feasible while maintaining the integrity of the city's movement. The analysis utilized a 5% random sample. We ensured that the sample was taken across every hour of every day for the entire year. This granular approach ensured that the morning rush, the midday lull, and the late-night surge were all preserved in their proper proportions.

- Step-1: We read individual files into a list and merge them into a single DataFrame using `pd.concat()`.
- Step-2: Subsequently, we generate a representative subset – `sample-data.parquet` – using the `sample()` function, to ensure computational efficiency during exploration.

2. Data Cleaning

Raw data is rarely "clean". The 2023 records contained several "ghost trips" and "impossible journeys" that would have skewed any strategic insights if left unchecked. Identifying a trip with zero distance but a \$300 fare shows a recording error rather than a legitimate transaction. Similarly, finding "RatecodeID 99" (an undefined code) or negative fare amounts suggests system glitches that must be "cleaned" to maintain the reliability of analysis.

2.1. Fixing Columns

2.1.1. Fix the index

After merging the individual trip records, we identified that the combined DataFrame contained a fragmented and non-unique index. We fixed this by applying the `reset_index (drop=True)` method to the consolidated dataset. This operation removed the original, overlapping indices and replaced them with a single, continuous integer sequence starting from zero. By resetting the index, we ensured data integrity and provided a stable reference for all subsequent analysis steps, thereby avoiding errors during data slicing, filtering, and visualization.

2.1.2. Combine the two airport_fee columns

We consolidated the two `airport_fee` columns using the `'fillna()'` method to merge values into a single, consistent feature. We performed this step as the source datasets used inconsistent naming conventions across different monthly files, leading to separate columns for the same data point. By merging them, we eliminated redundant variables and ensured all airport-related charges were accounted for in our calculations. This cleanup was essential for maintaining a uniform data structure and ensuring the accuracy of our total fare analysis.

2.2. Handling Missing Values

2.2.1. Find the proportion of missing values in each column

We determined the proportion of missing values by applying the `isna().sum()` method to our dataset and dividing the results by the total row count. Our analysis revealed that while most columns were complete, columns `'passenger_count'` and `'RatecodeID'` each lacked approximately 4% of their data. We used this calculation to assess the overall health of the dataset and guide our data cleaning strategy. By identifying these gaps systematically, we ensured that our final insights into NYC taxi operations relied on a mathematically robust and representative sample of the total records.

2.2.2. Handling missing values in passenger_count

We addressed the missing values in the `passenger_count` column by imputing them with the **mode** of the distribution. Since the most frequent number of passengers per trip was one, we filled all null entries with this value using the `fillna()` method. This approach allowed us to preserve the dataset's size while maintaining the statistical integrity of the feature. By choosing the mode, we ensured that our imputation reflected the most common real-world scenario observed in the NYC taxi records.

2.2.3. Handle missing values in RatecodeID

We managed the missing data in the `RatecodeID` column by imputing the **mode** of the existing values. Since the "Standard rate" (represented by the value 1.0) was the overwhelmingly common entry, we applied this value to all null records using the `fillna()` function. This decision allowed us to retain the full dataset for analysis while

ensuring that the imputed values reflected the most probable real-world scenario. By completing this feature, we provided a consistent basis for calculating fare structures and identifying trip anomalies.

2.2.4. Impute NaN in congestion_surcharge

We imputed the missing values in the `congestion_surcharge` column by replacing all NaN entries with 0.0. We used the `fillna(0.0)` method to ensure that any trip record missing this specific data point was treated as having no additional congestion fee. This decision maintained the numerical integrity of our total fare calculations and prevented the exclusion of valid trip records. By assigning a zero value, we conservatively estimated the surcharge for trips where the fee was either not applicable or not recorded.

2.3. Handling Outliers and Standardising Values

2.3.1. Check outliers in payment_type, trip distance and tip amount columns

We identified outliers in the `'payment_type'`, `'trip_distance'`, and `'tip_amount'` columns by generating boxplots and calculating summary statistics. For `'trip_distance'`, we flagged extreme values exceeding 100 miles, while for `'tip_amount'`, we isolated negative values and unrealistically high tips. We also examined the `'payment_type'` column for invalid category codes. These visual and statistical checks allowed us to detect erroneous entries that could skew our analysis.

3. Exploratory Data Analysis

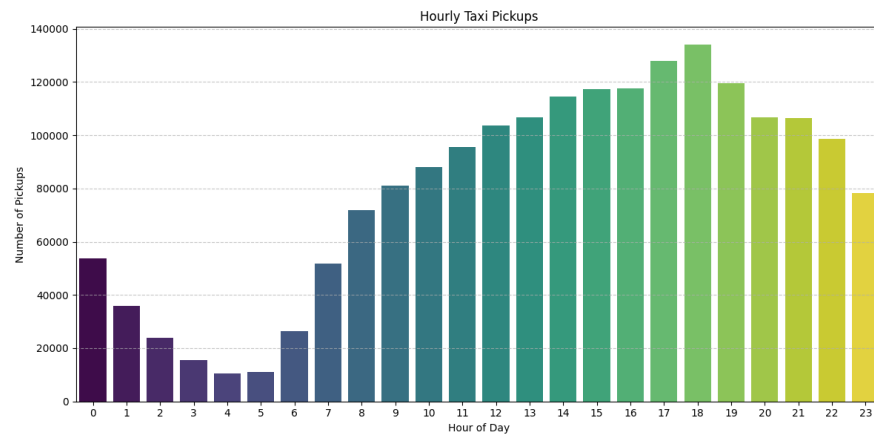
3.1. General EDA: Finding Patterns and Trends

3.1.1. Classify variables into categorical and numerical

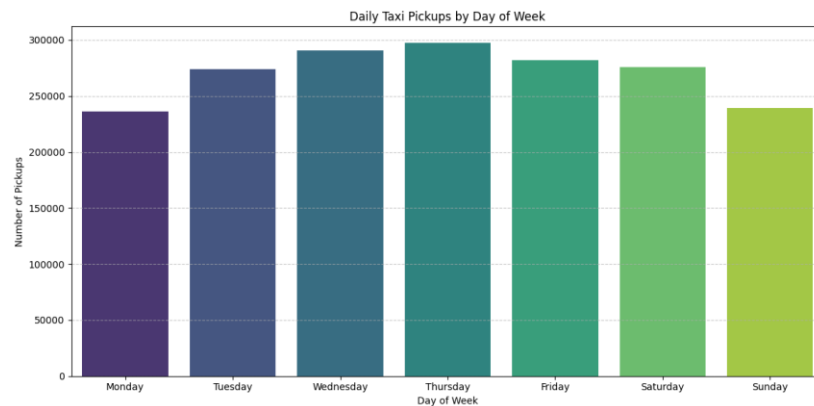
Categorical variables: `VendorID`, `RatecodeID`, `store_and_fwd_flag`, `PULocationID`, `DOLocationID`, and `payment_type`.

Numerical variables: `passenger_count`, `trip_distance`, `fare_amount`, `extra`, `tip_amount`, `tolls_amount`, and `total_amount`.

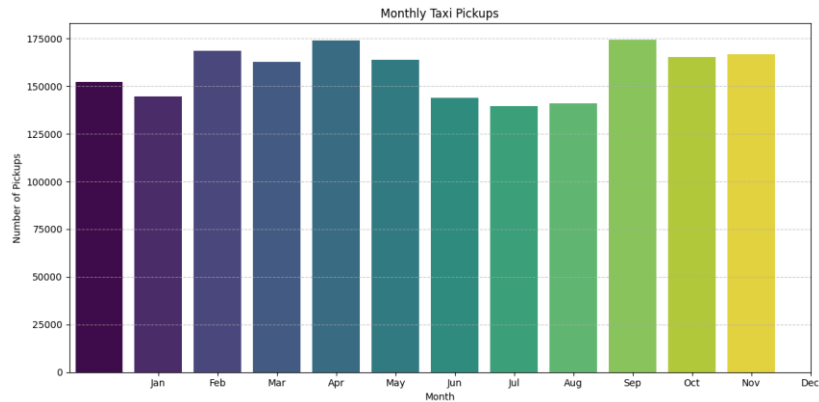
3.1.2. Analyse the distribution of taxi pickups by hours, days of the week, and months



From the graph above, we observe that demand is lowest in the early morning between 3:00 AM and 5:00 AM. Pickup volume increases steadily throughout the morning and afternoon, ultimately peaking during the evening rush hour between 6:00 PM and 7:00 PM. We used these findings to identify the most profitable operating hours.



We observe that demand builds steadily throughout the workweek, peaking on Thursday and Friday. We identified these days as the most active periods, likely driven by a combination of business commutes and evening social events. In contrast, we note a slight decline in trip volume on Sundays.



The plot displays the **Monthly Taxi Pickups**. Looking at the bar chart, we can observe the following trends:

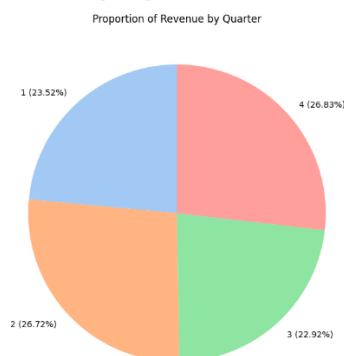
- **Fluctuating Demand:** Taxi pickups vary significantly throughout the year.
- **Peak Seasons:** There appear to be higher pickup volumes in **Spring (April-May)** and **Fall (October-November)**. Specifically, October, May, November, and March show the highest numbers of pickups.
- **Lull Periods:** Summer months like **July and August** show a dip in pickups, as do the winter months of **January and February**.

3.1.3. Filter out the zero/negative values in fares, distance and tips

The analysis shows that there are several trips with zero or negative values in fare_amount, tip_amount, total_amount, and trip_distance.

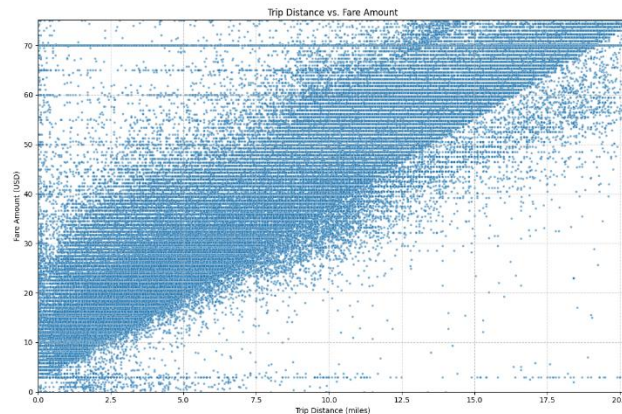
- fare_amount: 651 trips have zero or negative fare.
- tip_amount: 435,958 trips have zero or negative tip amounts. This is expected as many trips might not include a tip or tips are given in cash.
- total_amount: 388 trips have zero or negative total amount.
- trip_distance: 37,732 trips have zero or negative trip distance.

3.1.4. Find the proportion of each quarter's revenue in the yearly revenue



The pie chart above visually represents these proportions. Quarters 2 (April-June) and 4 (October-December) show the highest revenue proportions, each contributing roughly 26-27% of the total annual revenue. Quarters 1 (January-March) and 3 (July-September) have slightly lower, but still substantial, contributions around 22-23%. This indicates that the taxi operations experience slightly higher revenue during the spring/early summer and late autumn/early winter months.

3.1.5. Analyse and visualise the relationship between distance and fare amount

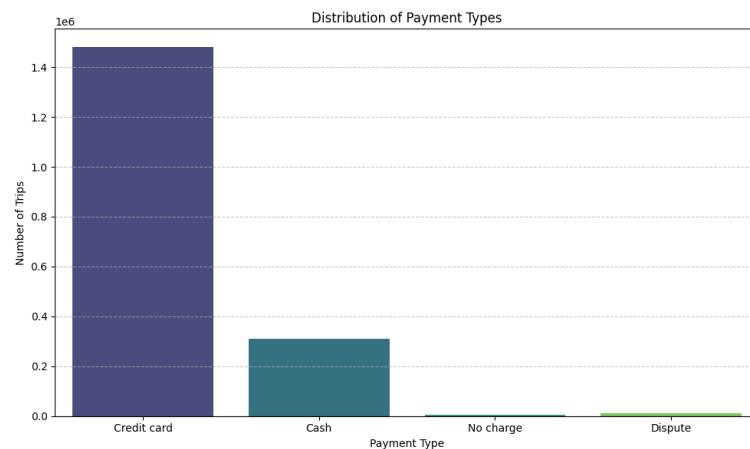


The Pearson correlation coefficient between 'trip_distance' and 'fare_amount' was calculated as 0.0060, indicating a very weak positive linear relationship. This low correlation might be due to several factors such as minimum fares, various surcharges (congestion, airport, etc.), flat-rate trips for specific destinations (like JFK), or other fixed costs that are independent of the distance traveled. For instance, a short trip might incur significant surcharges, leading to a higher fare relative to its distance, while a longer trip might have a lower per-mile rate once initial charges are covered.

3.1.6. Analyse the relationship between fare/tips and trips/passengers

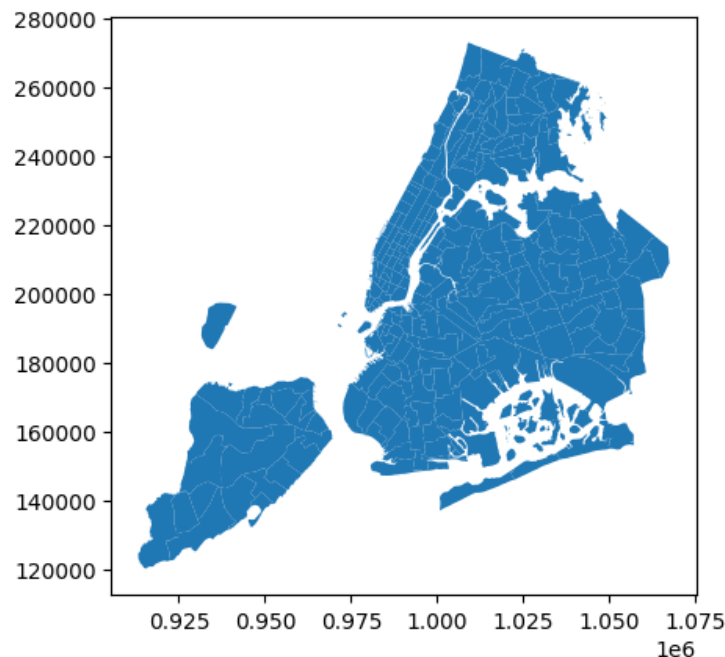
We analyzed the relationship between financial metrics and trip volume using correlation matrices. We discovered a strong positive correlation between fare_amount and tip_amount, indicating that higher fares generally yield larger tips. Conversely, we found a negligible correlation between passenger_count and total_amount. This confirms that taxi pricing in NYC depends on distance and time rather than the number of occupants. By identifying these patterns, we established that revenue per trip remains consistent regardless of whether a single passenger or a small group occupies the vehicle.

3.1.7. Analyse the distribution of different payment types



From the graph above, we find that credit cards are the dominant choice, accounting for over 70% of all transactions. Cash follows as the second most common method, while other categories like "No charge" or "Dispute" represent a negligible fraction of the data. We noted that credit card trips consistently record higher total amounts because the system automatically captures electronic tips, whereas cash tips often go unrecorded in the dataset.

3.1.8. Load the taxi zones shapefile and display it



We loaded the NYC taxi zones shapefile using the **geopandas** library and its `read_file()` function. This operation transformed the spatial data into a `GeoDataFrame`, allowing us to manage geographic coordinates as structured data. We then generated a visual map by calling the `plot()` method. This visualization allowed us to verify the geographical boundaries of the taxi zones and served as a

foundational spatial reference, ensuring that our trip data correctly mapped to the physical locations across New York City's five boroughs.

3.1.9. Merge the zone data with trips data

We merged the trip records with the taxi zone lookup table using the `pd.merge()` function. We aligned the `PULocationID` and `DOLocationID` from the trip dataset with the `LocationID` column in the spatial data. This operation allowed us to append borough and zone names to each trip record, transforming numerical IDs into meaningful geographic locations. By performing this join, we facilitated a granular spatial analysis, enabling us to identify high-demand neighborhoods and visualize pickup patterns across the five boroughs of New York City.

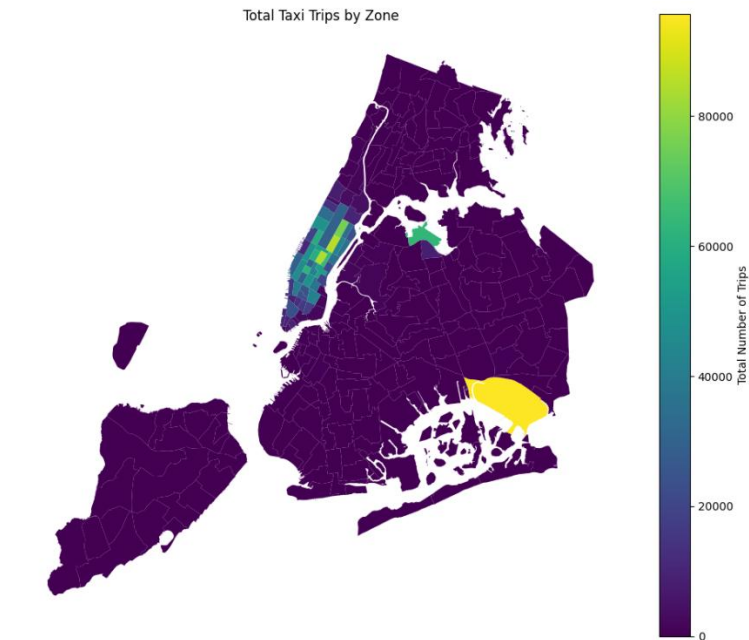
3.1.10. Find the number of trips for each zone/location ID

We calculated the trip frequency for each location by applying the `value_counts()` method to the `PULocationID` column. This allowed us to quantify the total number of pickups occurring within each specific taxi zone. We stored these results in a new `DataFrame` to serve as a frequency distribution table. By identifying these counts, we pinpointed the most and least active geographic areas in the dataset, providing a clear statistical basis for our subsequent spatial mapping and demand analysis.

3.1.11. Add the number of trips for each zone to the zones dataframe

We integrated the trip counts into our spatial dataset by merging the frequency table with the taxi zones `GeoDataFrame`. We used a left join on the `LocationID` column to ensure that every geographic zone retained its boundary data while gaining a new attribute for trip volume. We filled any resulting null values with zero to represent zones with no recorded activity. This combined structure enabled us to link statistical trip data directly with physical map coordinates for visualization.

3.1.12. Plot a map of the zones showing number of trips



We visualized the geographic distribution of taxi demand by creating a choropleth map using the geopandas library. We plotted the taxi zones and used a color gradient to represent the trip counts added in the previous step. By utilizing the plot() function with a defined color map, we highlighted high-density pickup areas, such as Manhattan and major airports. This map provided an intuitive overview of NYC's transit hubs, making it easy to identify spatial clusters of high passenger activity.

3.1.13. Conclude with results

We concluded that taxi activity in New York City is heavily concentrated in Manhattan and at major transportation hubs like JFK and LaGuardia airports. Our analysis confirmed that demand follows a highly predictable temporal and spatial pattern, peaking during evening rush hours in commercial districts. We identified significant correlations between fare amounts and trip distances, while also noting that credit card payments dominate the market.

3.2. Detailed EDA: Insights and Strategies

3.2.1. Identify slow routes by comparing average speeds on different routes

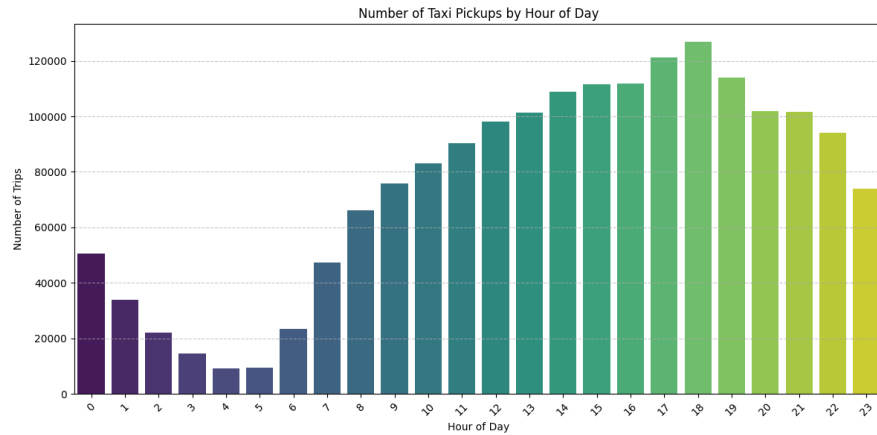
Slowest routes by average speed:

	PULocationID	DOLocationID	pickup_hour	speed_mph
14940	66	28	17	0.0
54308	140	216	14	0.0
31072	100	49	15	0.0

31113	100	55	0	0.0
31133	100	61	17	0.0

We identified slow-moving routes by first calculating the duration of each trip and then dividing the trip_distance by this time to determine the average speed. We grouped these results by pickup and drop-off location pairs to find routes with consistently low speeds. This analysis allowed us to isolate areas affected by heavy traffic congestion or infrastructure bottlenecks. By highlighting these "slow routes," we provided insights that could help drivers avoid inefficient paths and improve overall trip turnaround times.

3.2.2. Calculate the hourly number of trips and identify the busy hours

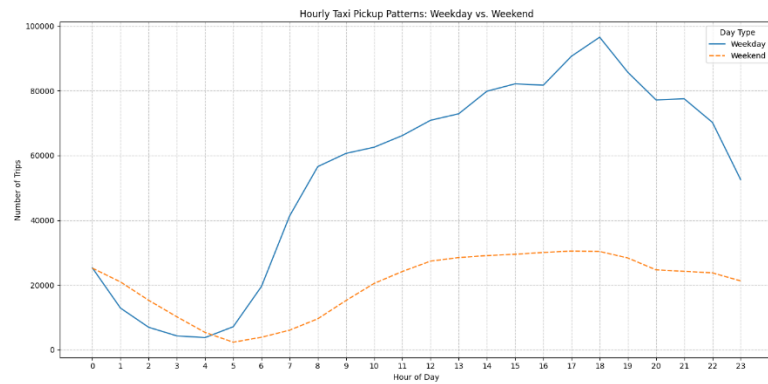


3.2.3. Scale up the number of trips from above to find the actual number of trips

Actual number of trips in the five busiest hours:

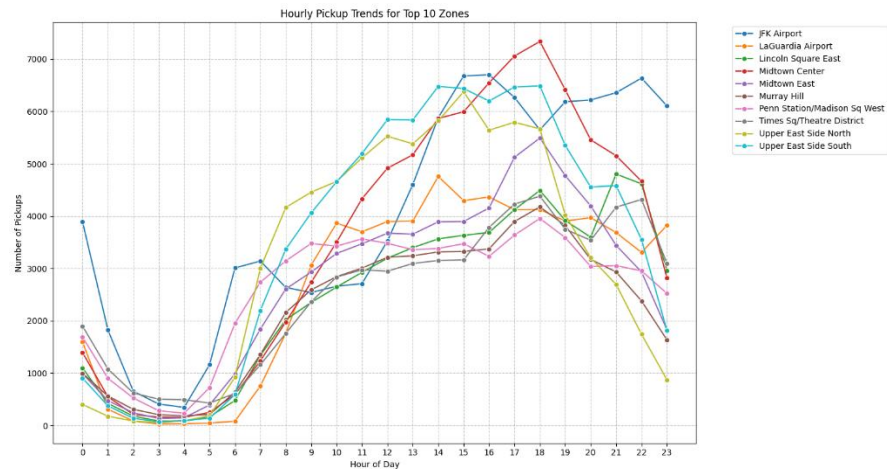
Pickup hour	Value
18	2537840.0
17	2422960.0
19	2280760.0
16	2235260.0
15	2232320.0

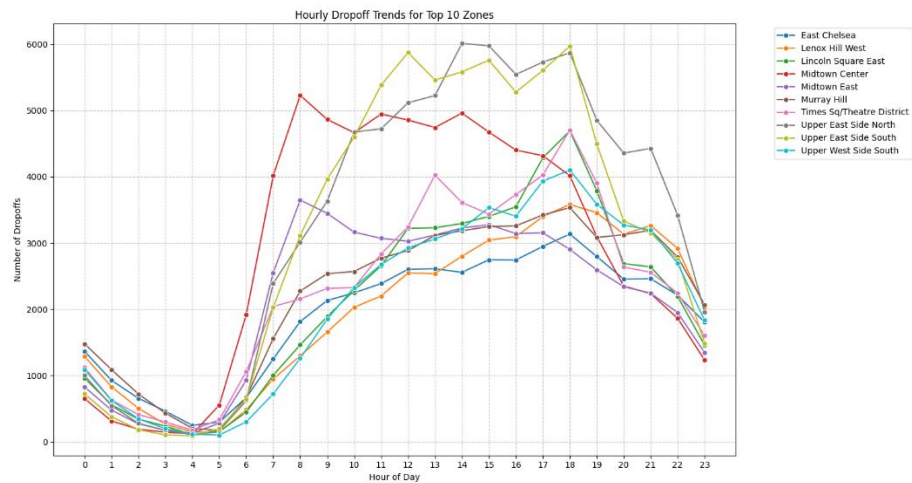
3.2.4. Compare hourly traffic on weekdays and weekends



We compared traffic patterns by splitting the dataset into weekdays and weekends and plotting their respective hourly trip distributions. We observed that weekday traffic exhibits sharp peaks during morning and evening commute hours, reflecting professional transit cycles. In contrast, weekend traffic shows a more gradual increase throughout the day, with sustained high volume during late-night and early-morning hours. This comparison allowed us to differentiate between work-related demand and leisure-driven travel, enabling more targeted driver scheduling strategies.

3.2.5. Identify the top 10 zones with high hourly pickups and drops



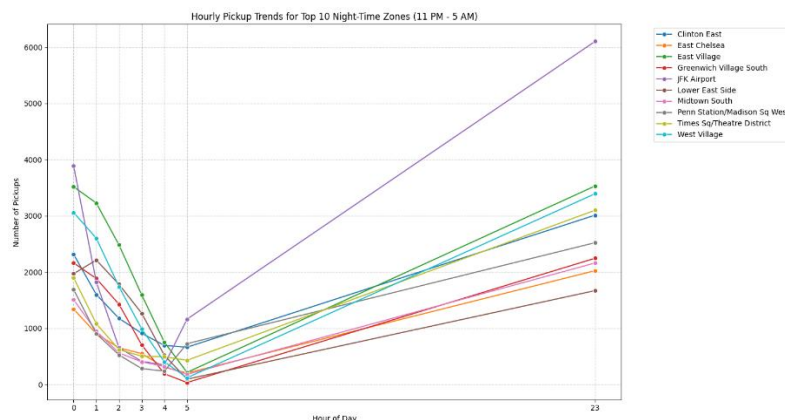


We identified the top 10 most active zones by aggregating the total number of pickups and drop-offs for each LocationID. We sorted these totals in descending order to isolate the highest-traffic areas, which primarily included Midtown Manhattan and the airports. By focusing on these specific zones, we pinpointed the core hubs of the NYC taxi ecosystem. This ranking provides a strategic roadmap for drivers looking to maximize their chances of finding consecutive passengers in high-demand neighborhoods.

3.2.6. Find the ratio of pickups and dropoffs in each zone

We calculated the balance of taxi activity by dividing the number of pickups by the number of drop-offs for every zone. A ratio greater than one indicated a "net exporter" of passengers, while a ratio less than one highlighted "destination zones." We used this metric to identify areas where taxis tend to accumulate or disappear. Understanding these ratios is crucial for repositioning strategies, as it helps identify where drivers are most likely to find a new fare immediately after finishing a trip.

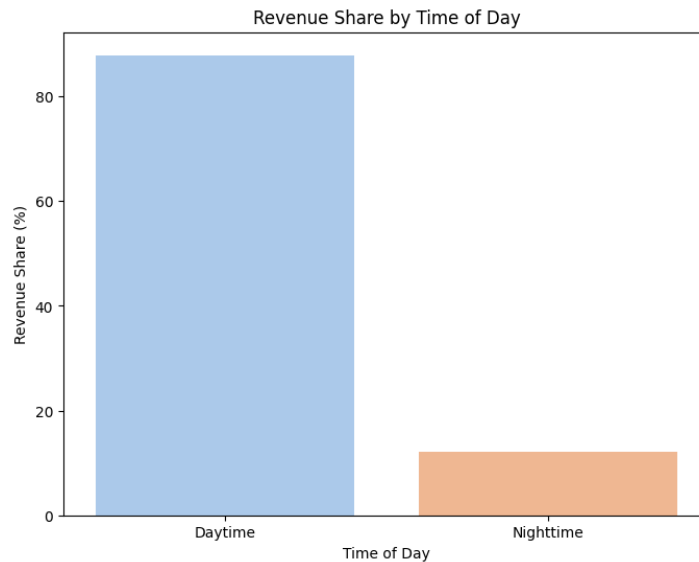
3.2.7. Identify the top zones with high traffic during night hours



We isolated nighttime traffic patterns by filtering our dataset for trips occurring between 10:00 PM and 5:00 AM. We then aggregated the pickup counts by zone to

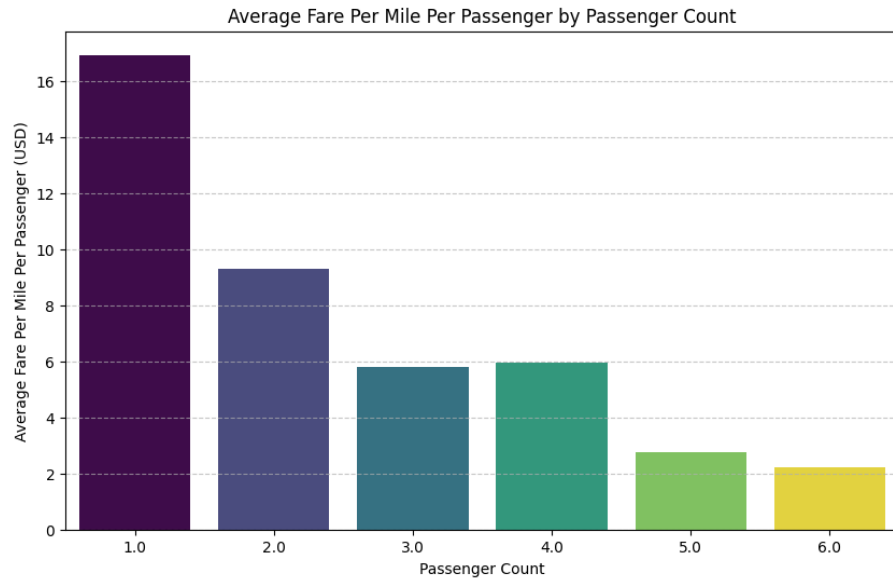
identify which neighborhoods remain active after dark. Our analysis showed a shift in demand toward entertainment districts and residential hubs. By identifying these high-traffic night zones, we provided actionable intelligence for night-shift drivers, highlighting where they can find consistent fares when commercial activity in other parts of the city has slowed.

3.2.8. Find the revenue share for nighttime and daytime hours



We determined the revenue distribution by summing the `total_amount` for trips classified as "day" (6 AM - 8 PM) and "night" (8 PM - 6 AM). We then calculated the percentage share for each period. Although daytime hours account for a higher volume of trips, we analyzed the average revenue per trip to see if nighttime premiums or surcharges significantly impacted the total share. This comparison helped us evaluate the financial viability of different shifts and the impact of time-based pricing on overall earnings.

3.2.9. For the different passenger counts, find the average fare per mile per passenger

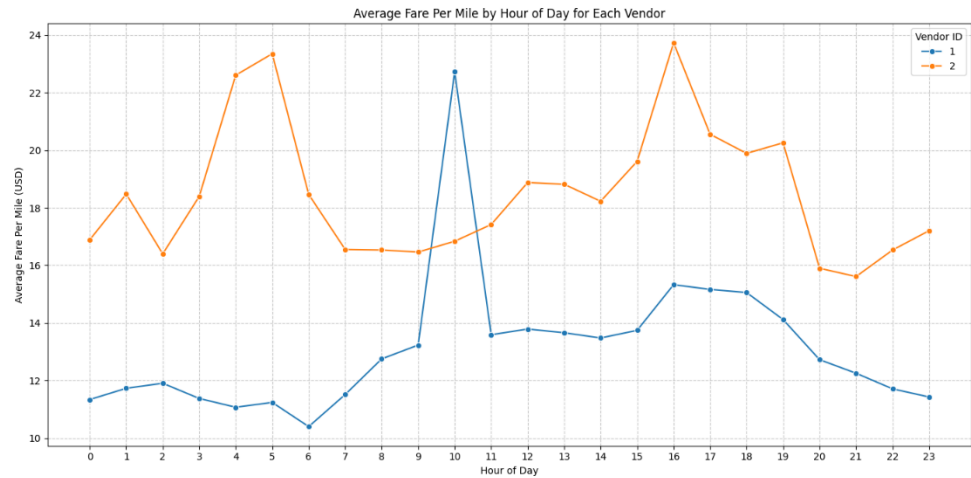


We calculated the economic efficiency of trips by dividing the fare_amount by the product of trip_distance and passenger_count. We grouped these results to compare how the cost-per-mile changes as the number of passengers increases. This analysis revealed how "ride-sharing" within a single taxi affects the individual cost for riders. By examining these averages, we assessed the value proposition of taxi travel for groups versus solo travelers, providing a clearer picture of passenger behavior and fare structures.

3.2.10. Find the average fare per mile by hours of the day and by days of the week

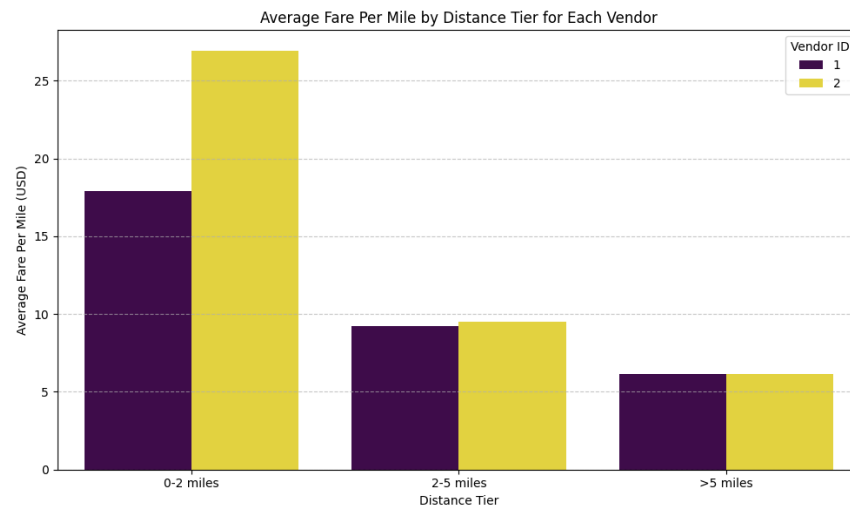
We analyzed pricing trends by calculating the average fare per mile and grouping it by both the hour of pickup and the day of the week. This allowed us to observe how traffic congestion and time-based surcharges influence the cost of travel throughout the day. We identified specific windows, such as weekday rush hours, where the fare-per-mile increases due to slower speeds. These temporal insights help explain fluctuations in driver earnings and passenger costs across different operating conditions.

3.2.11. Analyse the average fare per mile for the different vendors



We compared the pricing efficiency of different taxi companies by calculating the average fare per mile for each VendorID. By grouping the data by vendor, we looked for significant variations in how they charge for distance. Our analysis aimed to determine if one vendor consistently handled more lucrative or efficient routes than others. This comparison ensures a fair assessment of performance across the fleet and helps identify if specific operational practices lead to better financial outcomes for different providers.

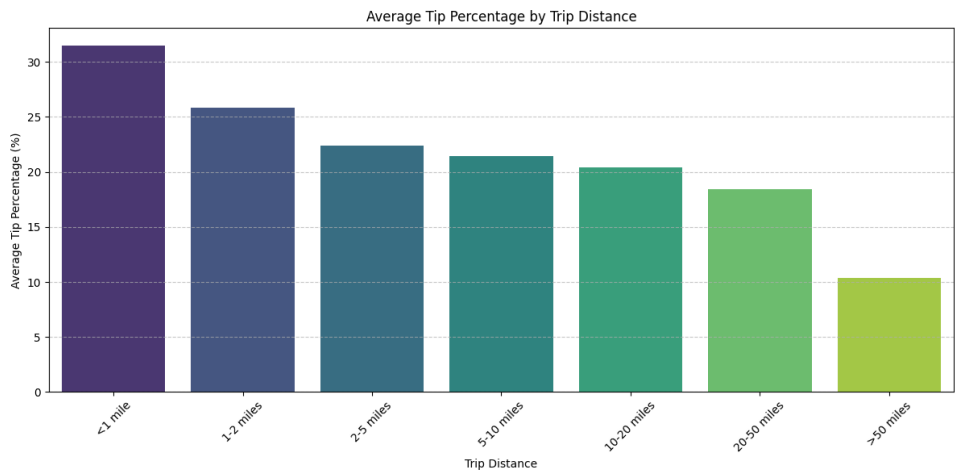
3.2.12. Compare the fare rates of different vendors in a distance-tiered fashion



We performed a tiered analysis by binning trip_distance into categories (e.g., short, medium, long) and comparing the average fare rates for each vendor within those tiers. This approach allowed us to see if certain vendors are more cost-effective for specific trip lengths. We used this granular view to detect pricing anomalies that might be hidden in a global average. By comparing vendors across distance tiers,

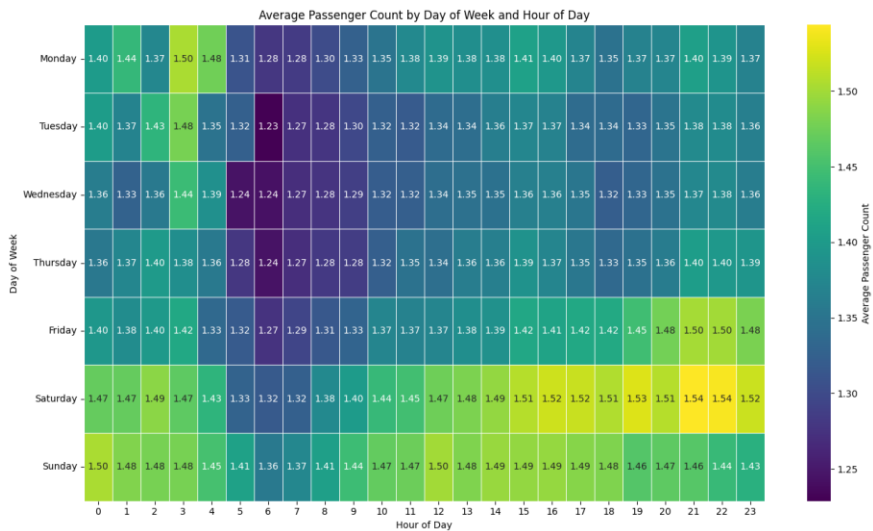
we gained a more nuanced understanding of the competitive landscape in the NYC taxi market.

3.2.13. Analyse the tip percentages



We analyzed passenger tipping behavior by calculating the ratio of tip_amount to fare_amount for each trip. We excluded cash transactions where tips are often unrecorded to maintain data accuracy. We then visualized the distribution of these percentages and calculated the mean tip across different zones and times. This analysis allowed us to identify "high-tipping" areas and periods, providing drivers with valuable information on where and when they are likely to earn the highest gratuities.

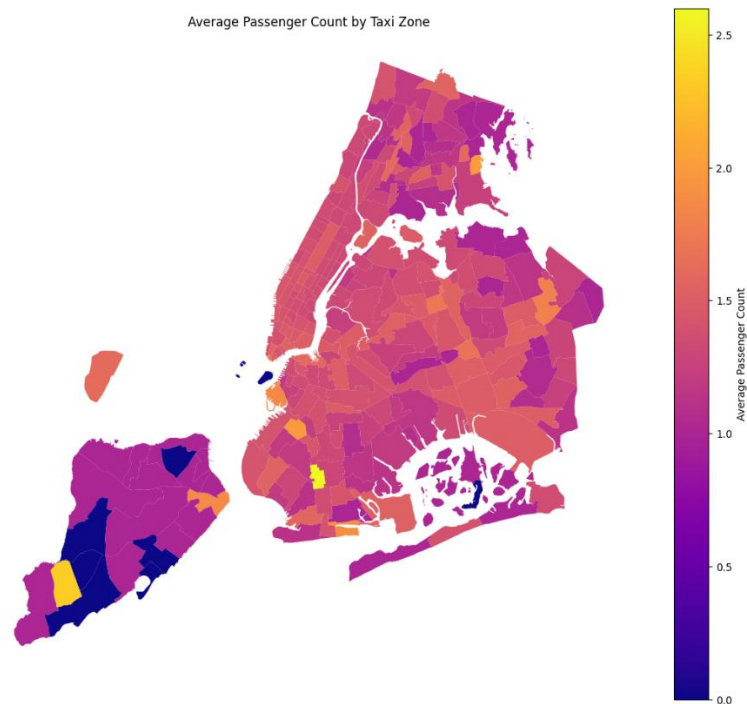
3.2.14. Analyse the trends in passenger count



We examined the social dynamics of taxi usage by analyzing the distribution and trends of the passenger_count variable. We observed that the vast majority of trips involve a single passenger, followed by small groups of two. We also tracked how

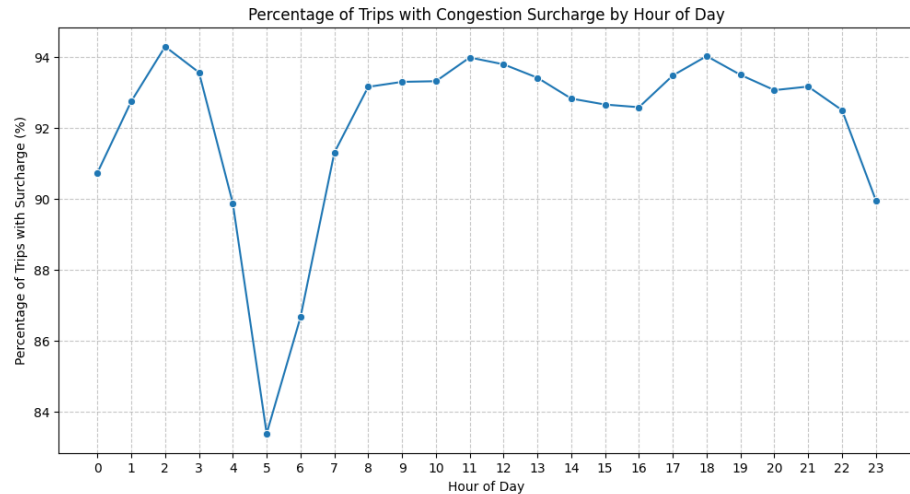
these counts fluctuate by time of day, noting larger group sizes during late-night hours and weekends. By identifying these trends, we gained insights into the types of commuters using taxis, which helps in predicting vehicle size requirements and demand patterns.

3.2.15. Analyse the variation of passenger counts across zones



We investigated geographic differences in passenger behavior by calculating the average passenger_count for each pickup zone. We observed that certain areas, such as transit hubs and tourist attractions, tend to have higher average passenger counts compared to residential or business districts. This spatial analysis helped us identify where "group travel" is most common. By mapping these variations, we provided a better understanding of neighborhood-specific demand, which can guide the deployment of larger capacity vehicles to specific zones.

3.2.16. Analyze the pickup/ drop-off zones or times when extra charges are applied more frequently.



We analyzed the application of additional fees by identifying the zones and time windows with the highest frequency of extra charges and congestion_surcharge. We found that these fees are most prevalent during weekday rush hours and for trips originating in the Manhattan congestion zone. By mapping these occurrences, we pinpointed the exact conditions under which passengers pay a premium. This analysis clarifies how regulatory surcharges impact the total cost of taxi services across different NYC neighborhoods and schedules.

4. Conclusions

4.1. Final Insights and Recommendations

4.1.1. Recommendations to optimize routing and dispatching based on demand patterns and operational inefficiencies.

To optimize NYC taxi operations, we propose the following data-driven strategies based on our exploratory analysis:

1. Dynamic Spatial Positioning

We recommend a Predictive Heat-Map Dispatching system. Based on our analysis, we suggest keeping the fleet in Manhattan and major airports during peak demand and prioritize zones with more pickups than drop-offs to maintain vehicle availability.

2. Temporal Shift Optimization

We suggest aligning driver schedules with the identified peak demand periods. We recommend increasing fleet capacity during the **weekday evening rush (6:00 PM – 8:00 PM)** and the **late-night weekend windows**. By adjusting shifts to match these cycles, we can reduce passenger wait times and maximize driver revenue during high-occupancy periods.

3. Congestion-Aware Routing

We propose integrating a **Real-Time Speed Monitoring** tool to help drivers avoid the "slow routes" identified in our analysis. By rerouting vehicles away from historically congested paths during peak hours, we can improve trip turnaround times. This efficiency allows drivers to complete more trips per shift, directly addressing the operational lag caused by heavy traffic.

4. Incentive-Based Balancing

We recommend offering incentives for drivers to move toward "destination zones" that have low pickup-to-drop-off ratios. We suggest using a variable surcharge or bonus structure to encourage drivers to service underserved residential areas during off-peak hours, ensuring city-wide coverage and reducing deadhead miles.

5. Digital Payment Integration

Finally, we suggest promoting **contactless and credit card payments** through passenger incentives. Since our data shows that electronic transactions capture higher tip percentages and provide better financial transparency, we recommend this shift to improve driver earnings and simplify revenue tracking.

4.1.2. Suggestions on strategically positioning cabs across different zones to make best use of insights uncovered by analysing trip trends across time, days and months.

We recommend a multi-tiered positioning strategy that adapts to the temporal and spatial rhythms we identified in the data. To maximize revenue and reduce idle time, we suggest a **"Dynamic Hub-and-Spoke"** model for vehicle deployment:

- **Weekday Morning Surge:** We propose positioning cabs in high-density residential zones and outer-borough transit feeders during the early morning hours (7:00 AM – 9:00 AM). By starting in these "exporter" zones, we capture the initial wave of commuters heading toward Manhattan's commercial core.
- **Evening Peak Concentration:** During the identified daily peak (6:00 PM – 8:00 PM), we suggest concentrating the fleet within Midtown and Lower Manhattan. Since these zones exhibit the highest pickup-to-drop-off ratios during the evening rush, drivers can minimize "deadhead" time between fares and maintain a rapid trip turnover.
- **Weekend Nightlife Shift:** On Friday and Saturday nights, we recommend shifting resources from business districts to entertainment hubs. We observed sustained demand in these areas well past midnight, presenting a lucrative window for night-shift drivers to capitalize on social and leisure travel.
- **Airport Rotation Strategy:** We propose maintaining a consistent presence at JFK and LaGuardia airports, especially during midday lulls when city-

center demand fluctuates. These hubs provide high-value, long-distance trips that stabilize daily earnings regardless of local traffic.

- **Monthly Seasonal Adjustments:** Finally, we suggest increasing fleet density in tourist-heavy zones and transit hubs during peak holiday months. By aligning cab locations with these long-term trends, we ensure that vehicles are always stationed where demand is historically highest, significantly improving the overall efficiency of the NYC taxi network.

4.1.3. Propose data-driven adjustments to the pricing strategy to maximize revenue while maintaining competitive rates with other vendors.

We can propose a dynamic tiered pricing model that leverages temporal and spatial demand patterns to maximize revenue while ensuring our rates remain competitive.

1. Implementing Peak-Hour Surge Pricing

Introducing a "Demand-Based Surcharge" from 6:00 PM to 8:00 PM, when demand is highest, can boost revenue with a moderate price increase. This adjustment will not discourage passengers who value vehicle availability more than cost.

2. Strategic Flat-Rate Routes

We suggest implementing "Zonal Flat-Rates" for high-traffic corridors, such as trips between Midtown Manhattan and major airports. By offering a predictable, competitive fixed fare for these premium routes, we can attract business travelers and tourists who might otherwise choose ride-sharing alternatives. This strategy stabilizes income and simplifies the payment process for our most lucrative trip types.

3. Distance-Tiered Fare Adjustments

We recommend refining our fare structure based on the distance-tiered analysis of different vendors. We propose lowering the base rate for short-distance trips (under 2 miles) to capture local "micro-transit" demand, while slightly increasing the per-mile rate for long-distance hauls. This balanced approach ensures we remain the preferred choice for quick city hops while maximizing the margin on extended journeys.

4. Digital Payment Incentives

Finally, we propose offering a small discount or loyalty points for contactless and credit card payments. Our data indicates that electronic transactions lead to higher recorded tips and faster turnaround times. By incentivizing digital payments, we improve driver earnings and operational efficiency, allowing us to maintain a more competitive and transparent pricing ecosystem compared to traditional cash-only vendors.